

Probability Measure

Lecture 2 Notes

Definition 2.1 — Sample Space

A sample space Ω is any non-empty set.

Example 2.1: The sample space contains “all the states of the following”:

- Coin flipping: $\Omega = \{H, T\}$
- Weather: $\{\text{hot, cold, mild}\} \times \{\text{wet, dry}\}$

Definition 2.2 — Probability Measure (STA 347 version)

For a sample space Ω , we say P is a **probability measure** if the following hold:

- (i) $\forall E \subseteq \Omega$, $0 \leq P(E) \leq 1$, that is $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$ $\leftarrow$ the power set
- (ii) $P(\Omega) = 1$
- (iii) $\forall E_1, E_2, \dots \subseteq \Omega$ such that $E_i \cap E_j = \emptyset$ ($\forall i \neq j$) (mutually exclusive)

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Remark: It may be impossible for such a P to exist. To define exactly which subsets we want P to satisfy this axiom, that will require more advanced mathematical theories. For this course, we just assume such a P is defined for all subsets of Ω .

Example 2.2: Probability of coin flipping.

$$\Omega = \{H, T\}, \quad \mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

$$P : \quad \emptyset \mapsto 0, \quad \{H\} \mapsto \frac{1}{2}, \quad \{T\} \mapsto \frac{1}{2}, \quad \{H, T\} \mapsto 1.$$

Example 2.3: The uniform / Lebesgue measure.

Let $\Omega = [L, U]$ for some $L < U \in \mathbb{R}$.

$$P([a, b]) = \frac{b - a}{U - L} \quad \text{when } L \leq a \leq b \leq U.$$

Proposition 2.3

The following properties are satisfied by any probability measure P :

1. $\forall E \subseteq \Omega, \quad P(E^c) = 1 - P(E)$
2. $P(\emptyset) = 0$
3. $\forall E, F \subseteq \Omega$ such that $E \subseteq F, \quad P(E) \leq P(F)$
4. $\forall E, F \subseteq \Omega, \quad P(E \cup F) = P(E) + P(F) - P(E \cap F)$
5. $\forall E, F \subseteq \Omega, \quad P(E \cap F) = P(E) + P(F) - P(E \cup F)$
6. $\forall E, F \subseteq \Omega, \quad P(F \setminus E) = P(F) - P(E \cap F)$

“big event, big probability”

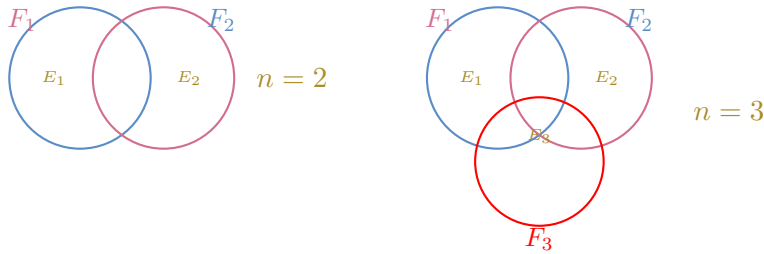
Proof via Def 1.2 (probability axioms).

Lemma 2.4 — Subadditivity

For a sequence $E_1, \dots, E_n \subseteq \Omega$,

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i).$$

Proof. We define $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$.



Then

$$\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i, \quad \text{and} \quad F_i \cap F_j = \emptyset \text{ if } i \neq j \quad (\text{mutually exclusive})$$

from the probability axioms. Since $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i$ and $F_i \cap F_j = \emptyset$ for $i \neq j$,

$$P\left(\bigcup_{i=1}^n E_i\right) = P\left(\bigcup_{i=1}^n F_i\right) = \sum_{i=1}^n P(F_i).$$

We also know $F_i \subseteq E_i$ (as $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$), so $P(F_i) \leq P(E_i)$. Hence

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(F_i) \leq \sum_{i=1}^n P(E_i). \quad \blacksquare$$

Lemma 2.5

Let P be the Lebesgue measure on $[0, 1]$. Then $P(E) = 0$ for any countable set E .